

Parametric emulation and inference in computationally expensive integrated urban water quality simulators

Antonio M. Moreno-Rodenas^{1,*}, Jeroen G. Langeveld^{1,2}, Francois H.L.R. Clemens^{1,3}

¹ Section Sanitary Engineering, Water Management Department, Faculty of Civil Engineering and Geosciences, Delft University of Technology, Delft, 2628 CN, The Netherlands

² Partners4UrbanWater, Javastraat 104A, Nijmegen 6524 MJ, The Netherlands

³ Department of Hydraulic Engineering, Deltares, Delft 2600 MH, The Netherlands

* Correspondence: a.m.morenorodenas@tudelft.nl

ABSTRACT

Simulating water quality dynamics in highly urbanised catchments involves the integrated modelling of several subsystems (e.g. wastewater treatment processes, urban drainage and receiving water dynamics). The complexity of these integrated catchment models grows fast, leading to potentially highly over-parameterised and computationally expensive model structures. The river water quantity and quality process parameter values become highly relevant when simulating pollution dynamics (e.g. dissolved oxygen depletion events). Applying uncalibrated, parametric prior knowledge extracted from literature or from experts in these models can lead to significant uncertainties in dissolved oxygen simulation. Inferring local parameter values from field observations is necessary for a meaningful usage of the integrated modelling platform. Unfortunately, the amount of data required to calibrate river parameters is often large, since individual, local storm events as well as seasonal dynamics should be captured jointly. Thus, computationally expensive simulators render the use of formal likelihood based inference schemes impractical. In this study, the emulation of dynamic dissolved oxygen concentrations at a large-scale integrated catchment model for an urbanised river catchment is presented. This rendered a significant acceleration of the model evaluation, thus allowing for the application of formal inference schemes or sensitivity analysis. Parameters for flow and dissolved oxygen dynamics are inferred in a formal Bayesian scheme. The use of different likelihood structures, homoscedastic, independent Gaussian, heteroscedastic independent Gaussian and autocorrelated processes is discussed. Additionally, a global sensitivity analysis provided a better understanding of the model and system's behaviour.

KEYWORDS

Bayesian inference, Integrated catchment modelling, Dissolved oxygen simulation, Emulation

1 INTRODUCTION

Integrated urban water quality modelling focuses on the joint simulation of processes affecting water dynamics through the urban-river system (Muschalla et al. 2009; Rauch et al. 2002). These models jointly evaluate wastewater treatment processes, urban drainage and river dynamics, which usually generates a rapid escalation of complexity (Benedetti et al. 2013). The representation of all subsystems involved, produces highly parameterised conceptualisations, requiring a large amount of data in the identification-calibration process (Langeveld et al. 2013a). Additionally, the dynamics of interest often occur at very different time-space scales, for instance, urban combined sewer overflow (CSO) discharges have a characteristic time of minutes-hours whereas river dissolved oxygen dynamics is at hourly-monthly scale. Quantifying and analysing uncertainties in these platforms is required to avoid over-confidence in modelling results and guide model improvement (Deletic et al. 2012; Tscheikner-Gratl et al. 2017). Also, many analysis strategies require sampling intensive techniques as parametric inference (Dotto et al. 2012). For instance, inference schemes often require a large number of model evaluations (on the order of 10^4 - 10^5) to reach convergence. This renders the inference in the original simulation platform impractical for most real-scale integrated catchment modelling studies, which might take between minutes to hours per sample.

One approach to speed up convergence time is the use of optimised sampling schemes; as drawing samples in parallel (Goodman and Weare 2010; Laloy and Vrugt 2012) or with informed adaptive MCMCs (Hoffman and Gelman 2014). However, this still often requires a prohibitive number of model samples (for practical

engineering standards). Another strategy is the use of data-driven or mechanistic model emulation, where a black-box mathematical representation is used to replicate the link between a vector of parameters and the dynamic response of the simulator. In this line, Laloy et al. (2013) proposed a two stage sampling scheme, first generating a rough estimate from a model emulator and later from the simulator itself to perform parametric inference in a groundwater model. Carbajal et al. (2017) compared the performance of mechanistic vs. data-driven emulation for urban drainage simulators, concluding that in general a data-driven approach is to be preferred unless confronted with highly sparse training datasets. Yang et al. (2018) used a Gaussian process data-driven emulator to perform a parametric uncertainty analysis in a semi-distributed hydrological model. Those approaches are useful when the dimensionality of the problem allows for the fit of a data-driven emulator. Yet the use of dynamic input variability remains a challenge, limiting the use of emulators to represent the link of mainly global parameters under a fixed time-window. Moreno-Rodenas et al. (2018) presented a data-driven emulator scheme for a physically based flooding model to represent together rainfall dynamic input and model parametric variability. This allows for including input and parameters in uncertainty propagation schemes for arbitrarily long spatially homogeneous rainfall series. However, the generalisation of the input-parametric emulation strategy to other processes as water quality dynamics still remains unaddressed.

This study deals with a large-scale urbanised catchment model (integrating wastewater treatment processes, urban drainage and receiving water dynamics) for the estimation of dissolved oxygen environmental impacts. Uncertainties in a priori expert-literature elicited parametric ranges for the river water quality-quantity processes are too large for the operation of the modelling study. Therefore a formal Bayesian inference scheme is used to update prior knowledge based on local flow and dissolved oxygen observations. The model is too computationally expensive to be directly used in the sampling scheme. Thus two emulation platforms are created to represent a mapping for four water quantity and eight water quality process parameters for river flow and dissolved oxygen dynamic time series. The training is performed under a given dynamic realisation of the system during the full year of 2012. The performance of the two emulators is validated under an independent dataset. The emulators are used to implement a global sensitivity analysis and an inference scheme under various likelihood function conceptualisations. The residual structure of the posterior mean sample is contrasted with the building hypotheses of the likelihood function. This work shows as the use of a dynamic emulator scheme can assist in gaining further knowledge in the sensitivity and probably distribution of process-driven model parameters in a large-scale water quality simulator.

2 MATERIALS AND METHODS

The integrated catchment model

The objective of this modelling study is to simulate dissolved oxygen dynamics in the Dommel river, located in the south of the Netherlands (Fig 1). This is a small stream with a discharge between 4-20 m³/s, flowing through a mild-sloped lowland area with an approximated length of 120 km. The river receives the discharge of ~200 combined sewer overflow structures (CSO) from several municipal urban drainage systems (connected area of ~4400 ha) and of a wastewater treatment plant (WWTP) of ~750,000 p.e. This renders acute and chronic oxygen depletion events. The integrated model accounts for the interaction between these three subsystems (river, urban drainage and WWTP). The river model is conceptualised as a tank-in-series hydrological scheme consisting in 65 well stirred stretch sections where the pollutant fluxes and transformation rates are computed (the conceptual scheme is presented in Annex A). The main river water quality processes are shown in Table 5 (Annex A). The WWTP was modelled through an a ASM2d scheme representing three biological lines with primary clarifiers, activated sludge tanks and secondary clarifiers with a total capacity of 26,000 m³/h and a controlled by-pass storm settling tank with a capacity of 9000 m³/h. Urban drainage flow was represented by 29 lumped rainfall-runoff and sewer transport schemes (Solvi 2006) and 30 spatial clusters of CSOs. Sewer water quality was obtained by an influent generator at the WWTP (Langeveld et al. 2017) and by an event mean concentration multiplier at the CSO-receiving water links. Further detail in the modelling platform can be found at Langeveld et al. (2013b). The main spatial and temporal scales dominating the integrated system are discussed in Moreno-Rodenas et al. (2017). The full-integrated model was implemented in the platform WEST (DHI).

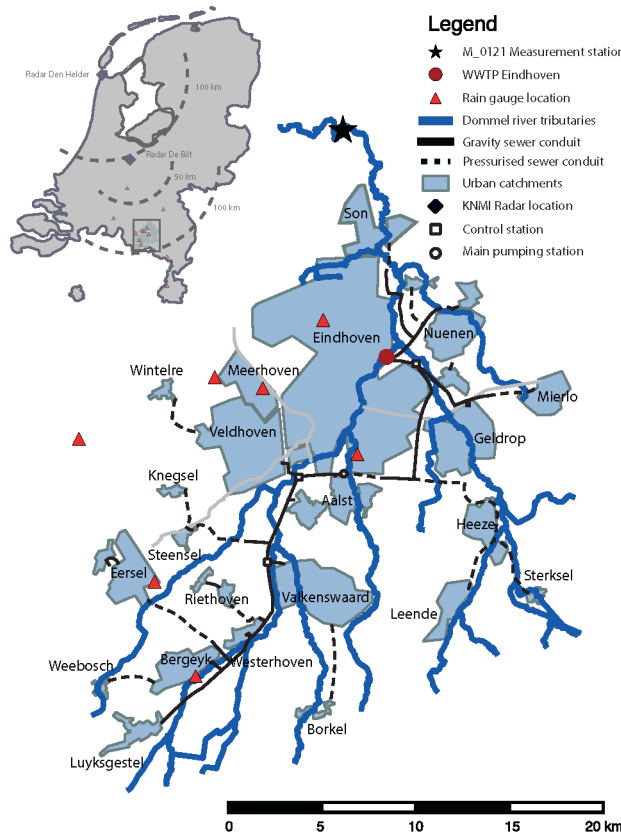


Fig 1 Urban water system of the Dommel.

Dynamic emulation

Two polynomial chaos expansion (PCE) structures (Xiu 2010) were used to emulate flow and dissolved oxygen dynamic series during one year (01-01-2012 – 31-12-2012). Firstly a training database of 200 samples was created by drawing realisations of flow series using a Latin hypercube sampling scheme (LHS) for four static parameters. Table 1 presents the river flow parameters along with the parametric range used to create the PCE database.

Table 1 River hydrology parameter PCE training ranges (emulation) and prior distributions (inference)

Name	Units	Description	PCE training	Prior distribution
n	$s \cdot m^{-1/3}$	Manning's roughness	$\sim U(0.02, 0.15)^*$	$\sim U(0.025, 0.12)$
K_z	-	Embankment slope multiplier	$\sim U(0.3, 2)$	$\sim U(0.7, 1.3)$
K_W_b	-	River bed width multiplier	$\sim U(0.3, 2)$	$\sim U(0.5, 1.5)$
K_h	-	Rural flow input multiplier	$\sim U(0.3, 2)$	$\sim U(0.7, 1.3)$

* $\sim U(a, b)$ refers to uniformly distributed probability density function between a and b.

A second emulator was built by simulating dissolved oxygen dynamics with the full-integrated model under the mean inferred flow parametric values. A LHS scheme was used to draw 200 samples from an eight-dimension parameter space of river water quality process parameters (Table 2). The process equations related to these parameters are shown at Annex A. Both training samples were taken by fixing the rest of parameters and input stochastic processes of the integrated model to their mean value.

Hermite orthogonal polynomials of order three were used in both emulators (Gautschi 1994). Training was done by a non-intrusive least squares fitting of the polynomial coefficients (Hadigol and Doostan 2018) using the chaospy implementation (Feinberg and Langtangen 2015). A detailed description of the polynomial expansion construction and fitting methodology used can be found at Moreno-Rodenas et al. (2018). An independent

validation set was created using a LHS of 100 samples for flow and 50 for dissolved oxygen. Nash-Sutcliffe efficiency was computed between the emulated and simulated one-year time series to validate the emulator's performance.

Table 2 River dissolved oxygen parameter PCE training ranges (emulation) and prior distributions (inference)

Name	Units	Description	PCE training	Prior distribution
Kd1	d ⁻¹	Decay rate for BOD fast	~U(0.3, 1)	~U(0.3, 0.8)
Kd2	d ⁻¹	Decay rate for BOD slow	~U(0.2, 1)	~U(0.2, 0.4)
Vs1	m · d ⁻¹	Sedimentation rate for BOD fast	~U(0.2, 40)	~U(0.5, 20)
Vs2	m · d ⁻¹	Sedimentation rate for BOD slow	~U(5, 100)	~U(10, 60)
TKd	-	Temperature coefficient for BOD oxidation	~U(1, 1.1)	~U(1.03, 1.1)
TKL	-	Temperature coefficient for reaeration	~U(1, 1.1)	~U(1, 1.03)
TSOD	-	Temperature coefficient for SOD	~U(1, 1.1)	~U(1.045, 1.1)
VKL	-	Velocity reaeration coefficient	~U(2, 8)	~U(2, 5)

Parametric Inference

Prior knowledge was encoded by means of range-constrained uniform probability density functions. Ranges were defined based on literature values and expert criteria (non-formal elicitation). Table 1 and Table 2 show the prior probability density function selected for each parameter.

This prior knowledge was updated by using an observation layout $Y_{obs} \in \mathbb{R}^{1 \times n}$ for hourly measured flow and dissolved oxygen concentration at the outlet of the Dommel catchment during a period of approximately seven months (15-Jan-2012 – 05-Aug-2012). The basic model-observation layout was defined as:

$$Y_{obs} = M(\mathbf{x}_0, \mathbf{x}, \boldsymbol{\theta}_M, \boldsymbol{\theta}_I) + Z \quad (1)$$

where M refers to the system of differential equations representing the integrated model (urban drainage, WWTP and receiving water body), which depend in a set of initial conditions $\mathbf{x}_0$ (computed by a warming up simulation period of 1 year for the WWTP initial conditions and a dedicated general initialisation of the previous 15 days between 01-Jan-2012 until 15-Jan-2012 for the rest of the variables), which are fixed in the sampling scheme. A set of dynamic inputs $\mathbf{x}(t)$, which contain, distributed urban rainfall spatial estimations, measured river and WWTP water temperature, river solar radiation, boundary conditions in river tributaries and control strategies. Many of those inputs contain substantial uncertainties due to the nature of the estimation-measurement process. A dedicated stochastic representation for each input was performed in a previous study, and the average expected realisation was used to simulate these training series. $\boldsymbol{\theta}_M$ represents a vector of m global model parameters, which were fitted by submodel calibration or expert elicitation. Whereas $\boldsymbol{\theta}_I$ represents the d model parameters used in the inference scheme for the river flow and dissolved oxygen dynamics (which are described in Table 1 and Table 2). In the contrast between measured (Y_{obs}) and modelled $M(\mathbf{x}_0, \mathbf{x}, \boldsymbol{\theta}_M, \boldsymbol{\theta}_I)$, a certain error (Z) is expected (Equation 1). The structure of this residual series Z is assumed a priori and validated once the posterior samples are performed. In the first try, the error generating process was assumed to be independent and identically gaussianly distributed. This assumption leads to a log-likelihood function with a reduced form:

$$\mathcal{L}(Y_{obs} | \boldsymbol{\theta}_I) \propto -\frac{1}{2} \log(\det(\boldsymbol{\Sigma})) - \frac{1}{2} (Y_{obs} - M(\boldsymbol{\theta}_I))^T \cdot \boldsymbol{\Sigma}^{-1} \cdot (Y_{obs} - M(\boldsymbol{\theta}_I)), \quad (2)$$

where $\boldsymbol{\Sigma}$ represents the residual covariance function, in this case:

$$\boldsymbol{\Sigma}_Z = \sigma_1^2 \cdot \mathbf{I} \quad (3)$$

being $\mathbf{I} \in \mathbb{R}^{n \times n}$ the identity matrix and σ_1^2 the constant variance of the residual structure, which is considered an hyperparameter of the inference and will also be updated.

Flow dynamics are also known to render heteroscedastic residual structures. This implies that residuals trend to be systematically larger when the flow is larger. This is represented by assuming that the error generating term Z does not have a constant standard deviation and this is a function of the variable value. Using a linear

relationship, the log-likelihood function has the same shape as in Equation 2, but the covariance matrix takes the form:

$$\Sigma_{Z_{het}} = (\sigma_1 + Q \cdot \sigma_2)^2 \cdot \mathbf{I} \quad (4)$$

where $\mathbf{I} \in \mathbb{R}^{n \times n}$ is the identity matrix, $Q \in \mathbb{R}^{1 \times n}$ is the computed variable vector, meanwhile σ_1 and σ_2 are the standard deviation hyperparameters of the error generating process, which are sampled from the posterior distribution.

Also, the inference of dynamic models often leads to autocorrelated residual structures. This is often taken into account by the use of a discrete autoregressive model of order p (Bates and Campbell 2001), or as formulated by Schoups and Vrugt (2010):

$$\Phi_p(B) \cdot z_t \sim N(0, \sigma_1) \quad (5)$$

being $\Phi_p(B) = 1 - \sum_{i=1}^p p_i \cdot z_{t-i}$ an autoregressive polynomial of order p for the residual z_t , with Gaussian updates.

An equivalent formulation to account for a correlation structure is discussed in (Honti et al. 2013), which used a bias description stochastic process B along with the error generating process. If assuming a stationary continuous constant bias and heteroscedastic residuals, Equation 2 defines the log-likelihood function, with a covariance matrix defined as:

$$\Sigma_{(Z_{het+B})_{ij}} = (\sigma_1 + Q(t_i) \cdot \sigma_2)^2 \cdot \delta_{ij} + \sigma_3^2 \cdot e^{-|d_{i,j}| \cdot \tau^{-1}} \quad (6)$$

which are the i and j elements of the covariance matrix $\Sigma_{(Z_{het+B})} \in \mathbb{R}^{n \times n}$, with δ_{ij} , the Kronecker's delta, $Q(t_i)$ the expected flow (at time t_i), $d_{i,j}$ the distance in hours between i and j elements, σ_3 the standard deviation of the stationary bias, and τ an extra hyperparameter which drives a correlation exponential decay. Del Giudice et al. (2013) discusses, that in practice the model effect and the bias descriptors can have a poor identifiability, thus inferring both, model parameters and bias hyperparameters, often requires assigning strong priors to the latter.

Table 3 Error model hyperparameters for the different hypotheses

Hyperparameter	Units	Description	Prior distribution
<i>Flow i.i.d Gaussian</i>			
σ_1	m^3/s	Std1, stationary standard deviation error	$\sim U(0, 10)$
<i>Flow independent heteroscedastic Gaussian</i>			
σ_1	m^3/s	Std1, stationary standard deviation error	$\sim U(0, 10)$
σ_2	m^3/s	Std2, stationary standard deviation error	$\sim U(0, 10)$
<i>Flow AR(3) Gaussian updating</i>			
σ_1	m^3/s	Std1, stationary standard deviation error	$\sim U(0, 10)$
$p_{1,2,3}$	-	Autocorrelation coefficients p1, p2, p3	$\sim U(0, 1)$
<i>Flow Heteroscedastic normal error and exponentially correlated bias</i>			
σ_1	m^3/s	Std1, linear intercept standard deviation error	$\sim U(0, 10)$
σ_2	m^3/s	Std2, linear slope standard deviation error	$\sim U(0, 10)$
σ_3	m^3/s	Std3, bias standard deviation	$\sim U(0, 10)$
τ	h	Tau, bias correlation exponential decay	$\sim U(10, 80)$
<i>Dissolved oxygen i.i.d Gaussian</i>			
σ_1	mgO_2/l	Std1, stationary error standard deviation	$\sim U(0, 10)$

Posterior samples were created using a Metropolis-Hasting algorithm (Hastings 1970; Metropolis et al. 1953). The joint probability distributions for flow and dissolved parameters were updated by drawing 50,000 samples from the MCMC (25,000 burn-in, 5 thinning). The Bayesian sampling implementation was performed using the python probabilistic programming package pymc version 2.3.6 (Patil et al. 2010).

Evaluating the likelihood distribution when using the full bias-description term involves inverting a covariance matrix of size n . In this case this was prohibitively expensive when using the original measurement layout ($n=4892$). Thus, a shorter period was used to test the inference of the bias description (26-Jul-2012 – 14-Sep-2012). In this case, only 4,000 accepted samples were used (2,000 burn-in, 2 thinning).

3 RESULTS AND DISCUSSION

Dynamic emulation of flow and dissolved oxygen concentrations

Two emulator schemes were trained using an orthogonal polynomial expansion. The scheme for flow dynamics was trained using 250 samples of the four-dimensional parameter space. The emulator for the dissolved oxygen dynamics was fitted using 200 samples. Fig 2 shows both training datasets for flow and DO simulated time series at the downstream end of the studied section of the Dommel catchment (Sind-Oedenrode, M0121, Fig 1). Both schemes used a 3th degree expansion of the Hermite orthogonal polynomials, and the expansion coefficients were fitted using a non-intrusive approach (least-squares fitting). Fig 3 shows the Nash-Sutcliffe efficiency (NSE) at 100 samples of the parametric space, which were used for validation of the emulator performance. Fig 4 shows the same test performed at 50 samples drawn from the dissolved oxygen concentration emulation scheme. The performance of both emulator implementations is consistent across the parameter ranges and varies between 0.99-1 NSE. The observed performance during validation was considered sufficient for the application of the inference scheme in this study.

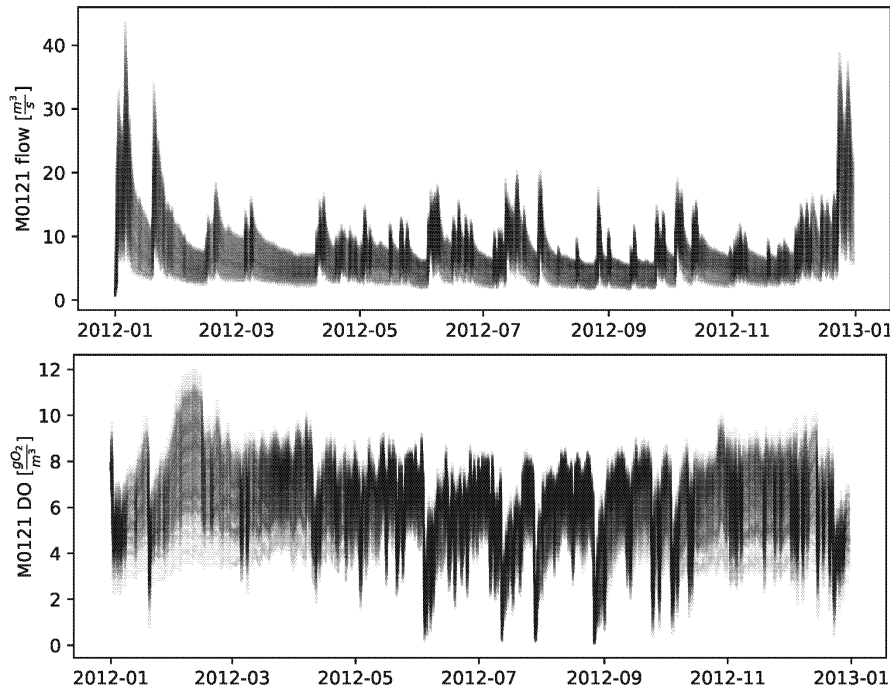


Fig 2 Training dataset for flow and dissolved oxygen at the outlet of the catchment (M0121)

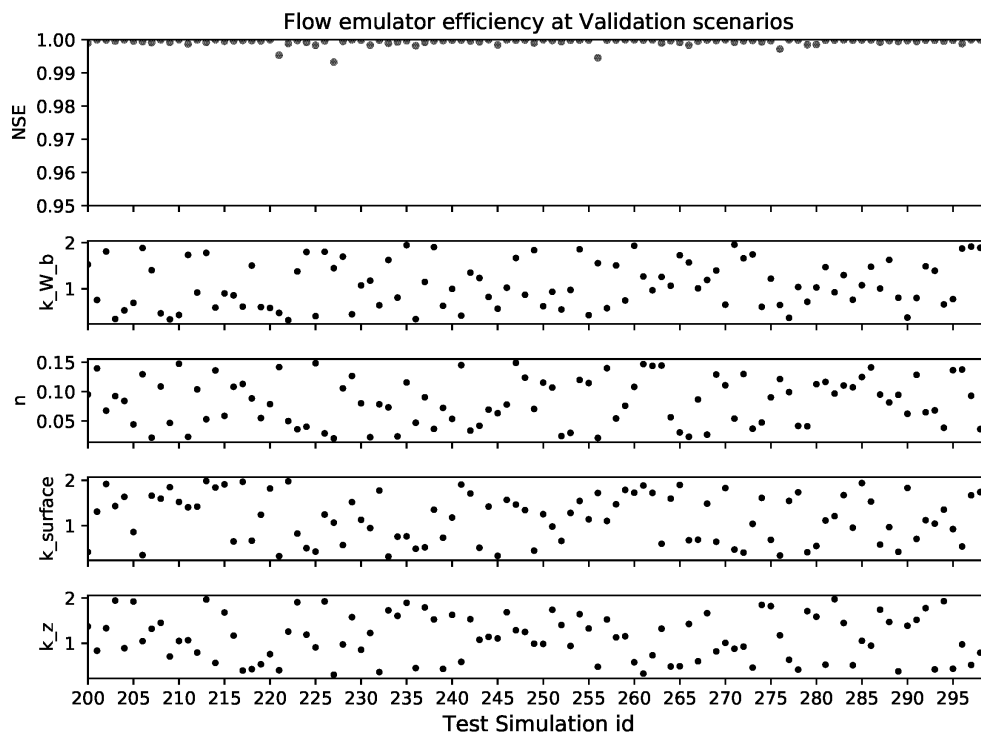


Fig 3 Nash-Sutcliffe efficiency at the flow emulator vs simulation for a four-dimensional parameter space under validation conditions.

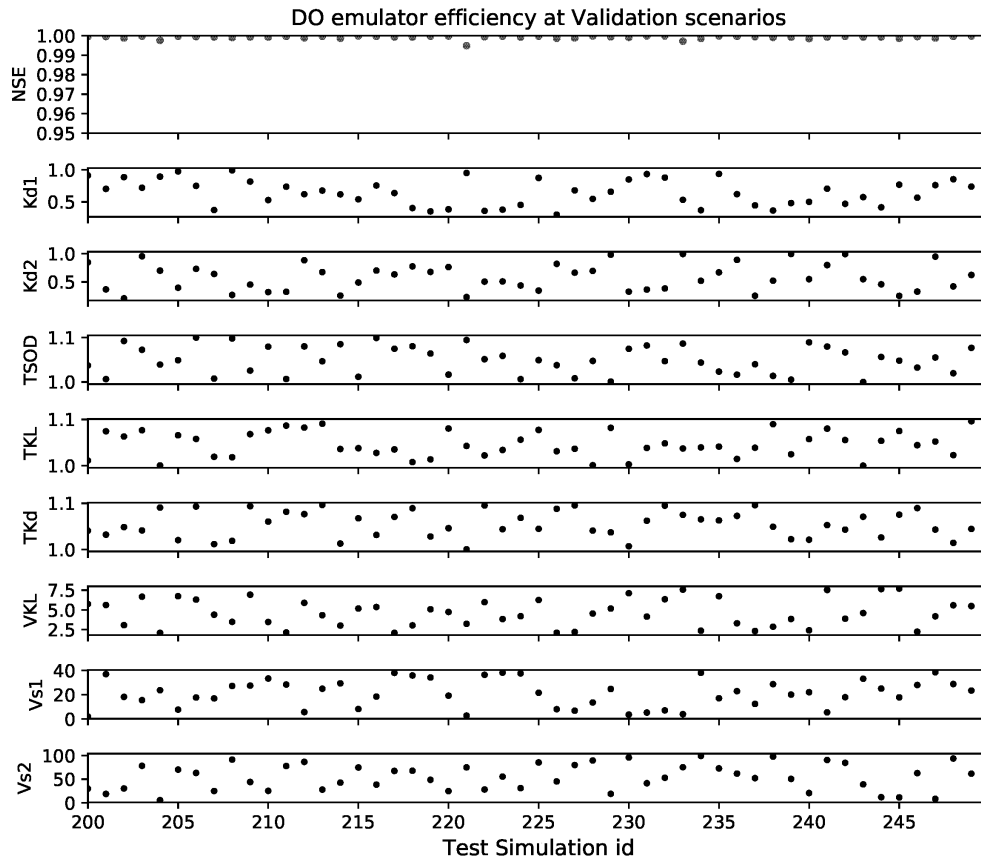


Fig 4 Nash-Sutcliffe Efficiency of dissolved oxygen emulator vs simulator for an eight-dimensional parameter space under validation conditions.

When increasing the dimensionality of the parametric space in the emulation scheme, the number of required training samples grows rapidly (Xiu and Karniadakis 2002). This causes that for some parametric space dimension, training the emulator becomes impractical, or to be a similar burden as directly sampling from the simulator. Table 4 presents the computational effort required to sample from the original simulator, training and operation of the emulator. In this case, those are the computed average timings when using a 2.2 GHz Intel Core i7 from mid 2014.

Table 4 Emulation vs. Model computational effort for one-year series (in seconds)

Phase	Flow	DO
Simulator sample (x1)	3300 s	3300 s
Training database (x200)	660×10^3 s	660×10^3 s
Orthogonal expansion fitting	14 s	61 s
Emulator sample	0.06 s	0.07 s

Global sensitivity analysis of process parameters

The fitted coefficients of the polynomial expansion have a direct interpretability in terms of the sensitivity of the different parameters. Yet, once validated, the emulator platform can be used to cheaply evaluate any combination of the parameters within the training ranges. Here, to facilitate comparison with other studies, both emulators were used to compute the Sobol global sensitivity analysis (Sobol 1993). The Sobol method is a widely applied variance-based global sensitivity metric, which identifies the contribution to the output's total variance of selected parameters or combinations. Fig 5 and Fig 6 depict the first order indexes from the prior distribution of parameters at the river flow and DO dynamics. Fig 5 show as the simulated flow level has a high dependency on the parameter $k_{surface}$ during dry-weather periods, whereas the manning's roughness (n) becomes relevant

during the rising limb of the hydrographs. k_{W_b} (multiplier for the river bed width) shows very little influence to the overall dynamics, meanwhile k_z (a multiplier for the slope of the embankment) has a similar, yet less pronounced effect when compared to hydraulic roughness.

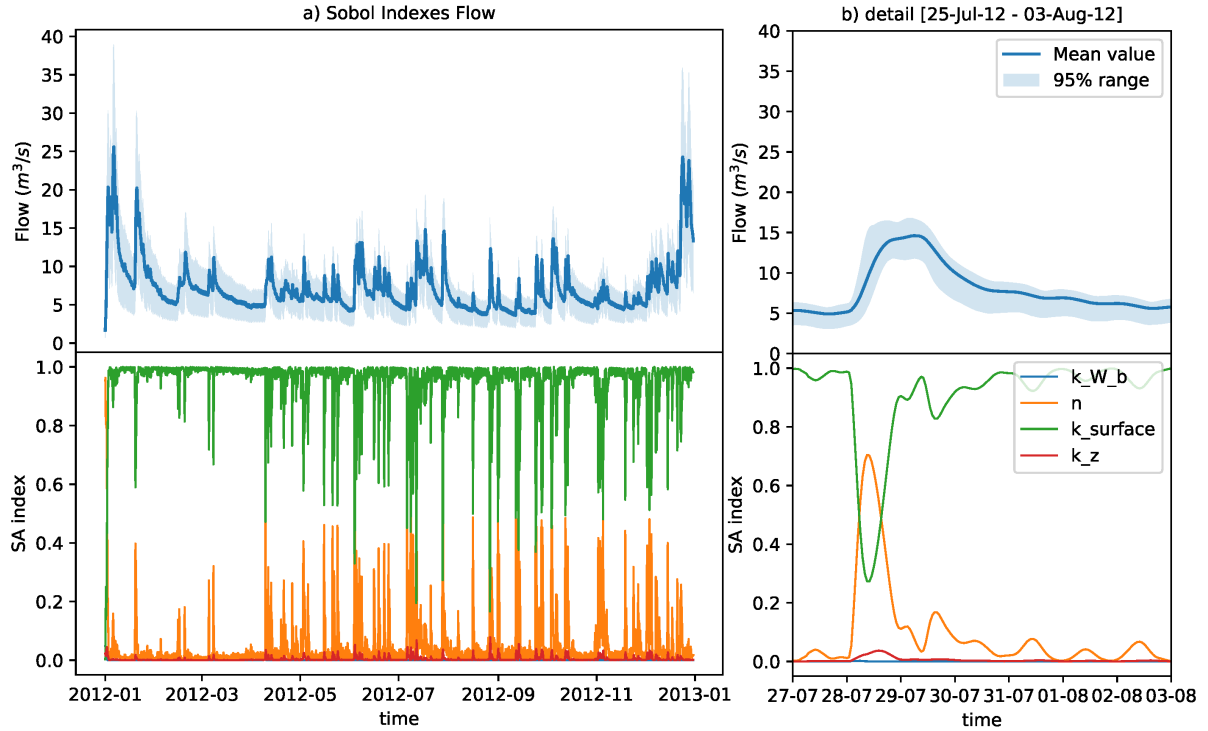


Fig 5 Sobol sensitivity indexes (first order) for the flow dynamics. Above, mean flow simulation and the 95% interval for the propagation of the parametric ranges. Below, sensitivity indexes for the four parameters. In the right (b) detail of the sensitivity during a medium-high intensity storm event.

The results of the study of the sensitivity for the DO concentration simulation are shown in Fig 6. The parameter controlling the reaeration rate (VKL) dominates the dry-weather DO variability during summer times. This influence decreases during winter, where the temperature coefficient for the sediment oxygen demand ($TSOD$) becomes increasingly relevant. This has to do with the temperature inhibition model structure, which influences the oxidation rate of organic matter for temperatures differing from 20 Celsius. During oxygen recovery patterns, $TSOD$ is also relatively relevant, since the main oxygen sink is the sediment layer. Sensitivity indexes do not show a consistent behaviour during acute oxygen depletion processes. Some depletion events, as the three occurring during July and September (also seen at Fig 6b) present as dominant parameters $kd1$ and $kd2$ which are the oxidation rates for the two fractions of suspended BOD in the system. However, the events occurred in June and the three in October showed to be more sensitive to a different parameter combination as TKL or $TSOD$, which are related to temperature driven reaeration or oxygen consumption (also slightly to $kd1$). This is a good example of the complexity of the underlying process, in which interactions are highly dependent in the dynamic state of the system. For instance, if a storm event activates predominantly northern CSOs (Fig 1), which are closer to the outlet of the catchment, there is less time for the degradation of suspended matter to occur, thus variations in the consumption rate might have a higher influence the process. Also, events in which the WWTP is the main source of discharge (and not the CSOs), the settling facilities of the WWTP might lead to a lower sediment build-up in the river and thus increasing the relevance of suspended organic matter degradation.

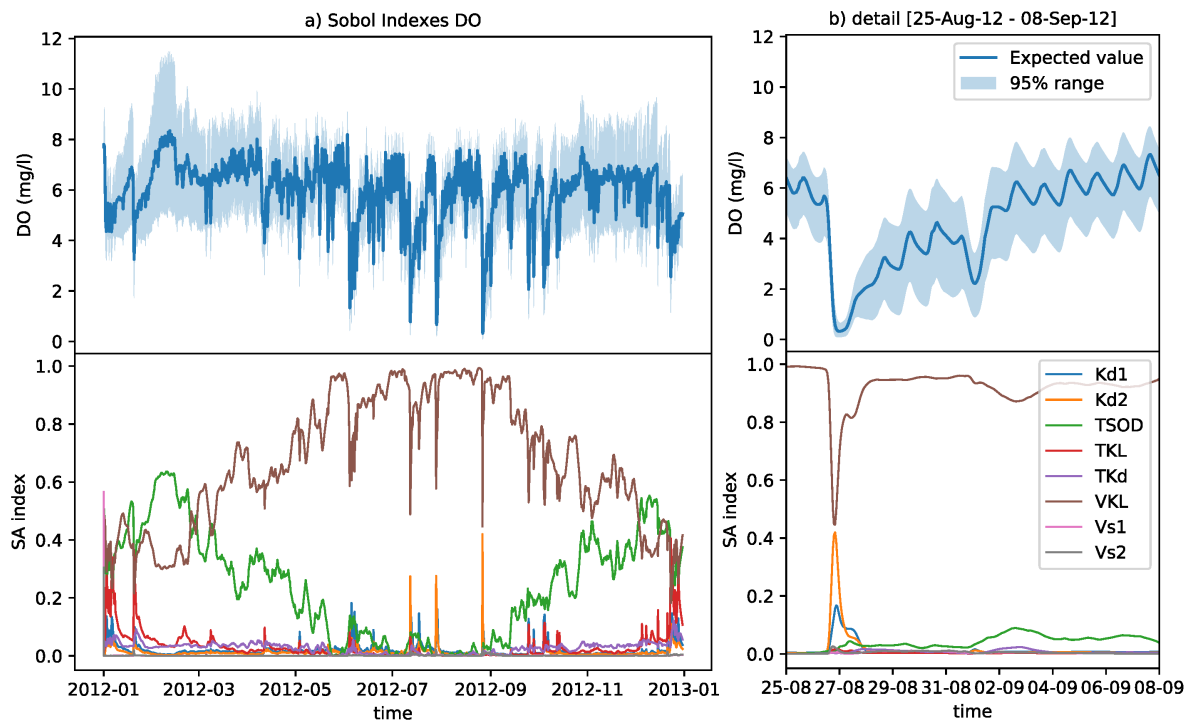


Fig 6 Sobol sensitivity indexes (first order) for the dissolved oxygen dynamics. Above, mean DO simulation and the 95% interval for the propagation of the parametric ranges. Below, sensitivity indexes for the eight parameters. In the right (b) detail of the sensitivity during a high intensity storm event.

Parametric inference

A data series of river flow and dissolved oxygen concentration (15-Jan-2012 until 05-Aug-2012) at M0121 station (the outlet of the catchment) was used to update prior knowledge (encoded as a probability distribution, Table 1 and Table 2). Fig 7 displays the comparison of measured flow (in black dashed line) and the posterior mean simulation (solid blue) plus its 95% uncertainty range. Also, a validation period (05-Aug-2012 until 31-Dec-2012) shows that the dynamics of the system are consistent. Flow series are reasonably well represented, although extremely dry periods induce a systematic overestimation of the flow as seen in July and in the beginning of September, this however is expected to have little influence in the water quality dynamics. The same comparison (measurement vs inferred and validation series) can be found in Fig 8 for the simulation of dissolved oxygen. The general dynamics of DO are captured, especially the depletion processes, daily and seasonal variation. Yet the uncertainty of the estimation is significantly large compared with the variable itself. This has certain implications for the applicability of the DO series in environmental studies. The use of DO deterministic estimations can lead to overconfidence on modelling results. During the 19th and the 29th of November 2012 Fig 8 shows a systematic mismatch of three acute depletion events, which are not captured by the model. This indicates that the model structure is apparently lacking some fundamental process, which has not been identified yet. The validated measuring data quality in this period is considered as good. This kind of depletion processes for the Dommel system should be further investigated in order to update the structure of the integrated catchment model in the future.

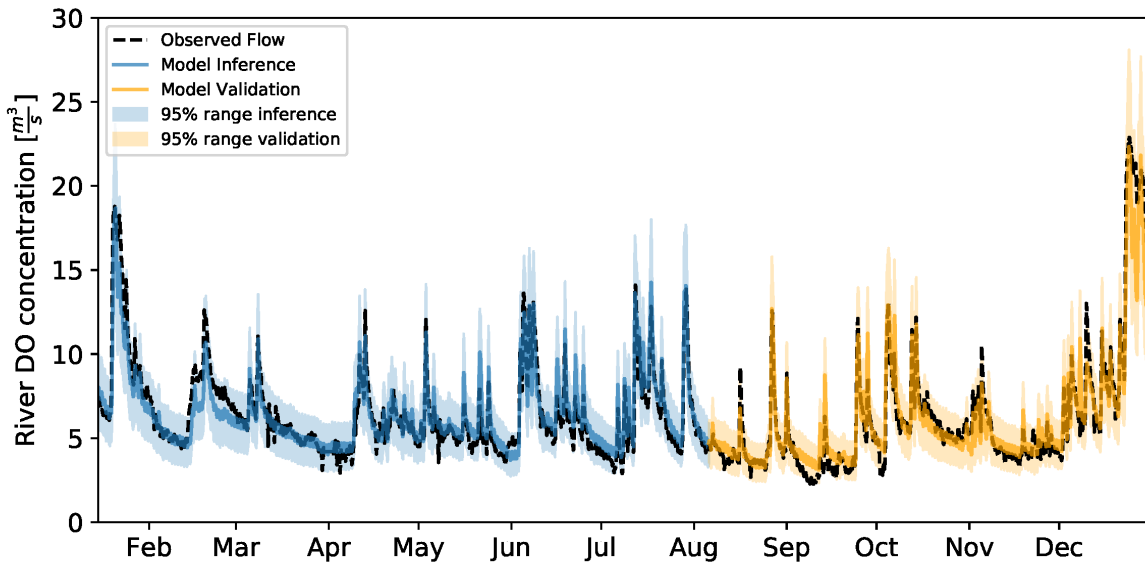


Fig 7 Posterior sample for the inferred flow dynamics between 15-Jan-2012 and 05-Aug-2012. In orange, the posterior distribution under validation conditions 05-Aug-2012 until 31-Dec-2012, in black observed flow at the station M0121.

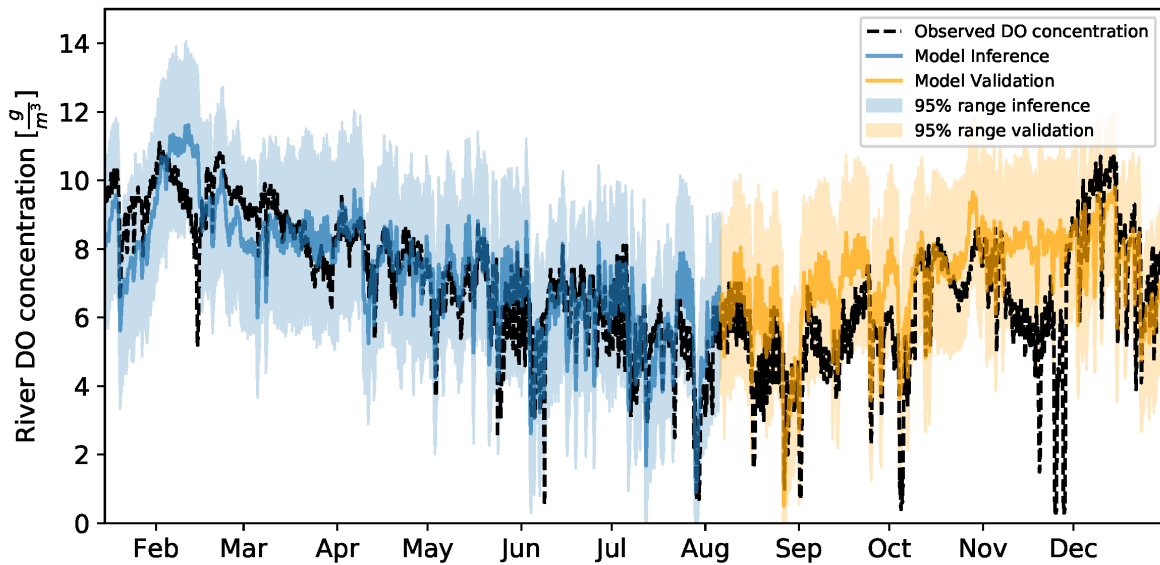


Fig 8 Posterior sample for the inferred dissolved oxygen dynamics between 15-Jan-2012 and 05-Aug-2012. In orange, the posterior distribution under validation conditions 05-Aug-2012 until 31-Dec-2012, in black observed flow at the station M0121.

The posterior probability density functions of the parameters for the water quantity and quality of the river section can be found at Fig 9 and Fig 10 respectively. The river variable k_{W_b} is poorly identified, which is denoted by the wide range of the posterior distribution (Fig 9, k_{W_b} diagonal). This is also supported by the very low sensitivity of this parameter to the overall flow dynamics (Fig 5). This parameter could have been eliminated during the inference phase, which would have accelerated the sampling of the posterior distribution by requiring fewer samples. Yet, since the use of the emulator allows for sampling at a very high speed (in the order of milliseconds) the addition of this parameter is not prohibitive. The rest of the parameters are appropriately identified, and they appear to be mostly mutually independent with the exception of a strong negative correlation between $k_{z_}$ and n . This implies that both elements interact in opposite directions, thus

partly compensating each other. The joint inference/calibration of both elements is therefore not recommended, since there is not enough information content in the objective variable to identify them independently. Further use of this model should therefore prioritize fitting n , since n has a much larger sensitivity than k_z .

Water quality variables show a mostly independent joint posterior distribution with the exception of $kd1$ and $kd2$, which show a mild negative correlation. This is explained by the fact that both parameters affect the same process at two fractions of BOD for which DO measurements are probably insufficient to discriminate.

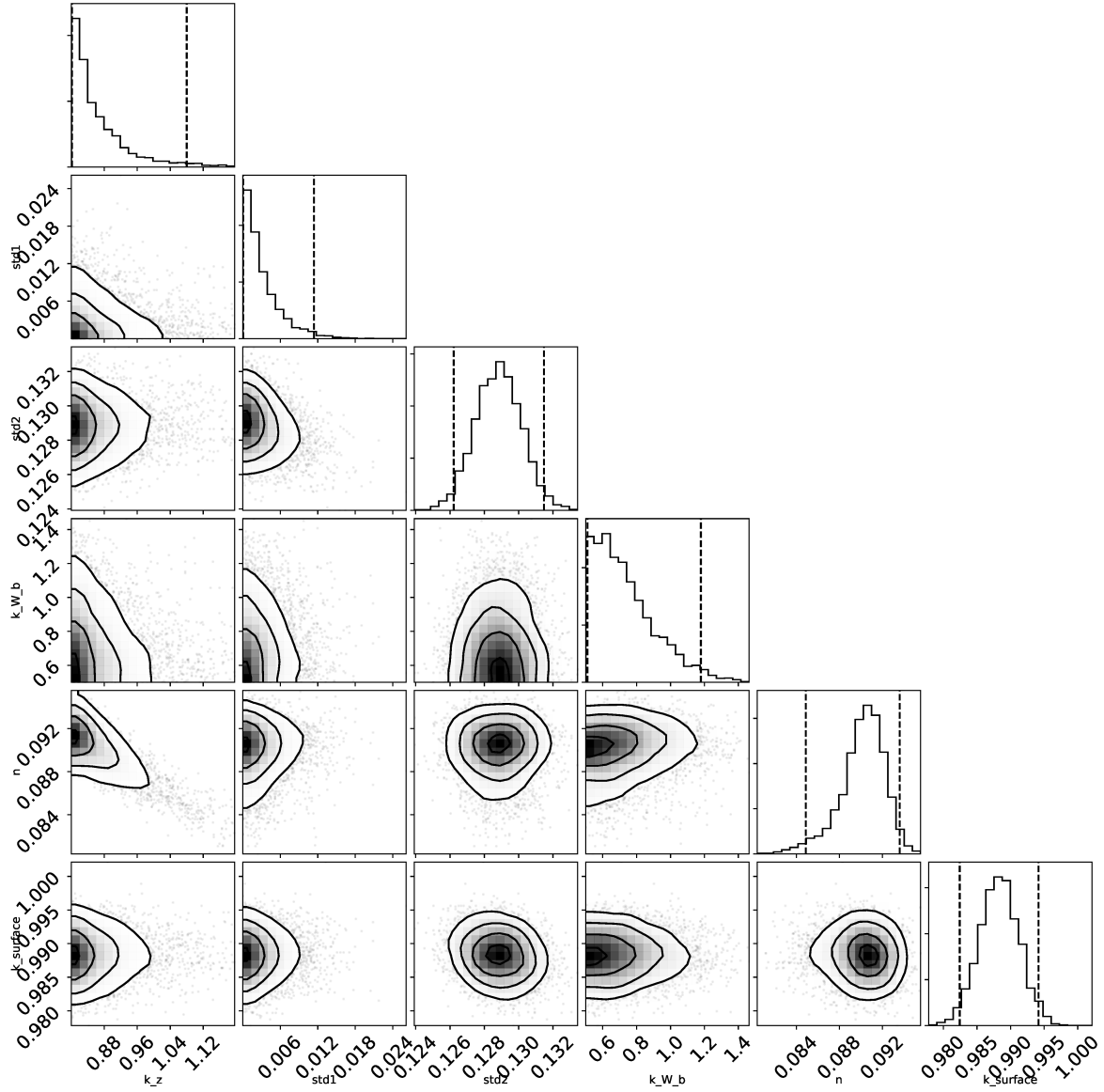


Fig 9 Posterior joint parametric distribution for the inference of the flow model parameters. Std1 and Std2 are hyperparameters of the selected error generation process (heteroscedastic, independent Gaussian)

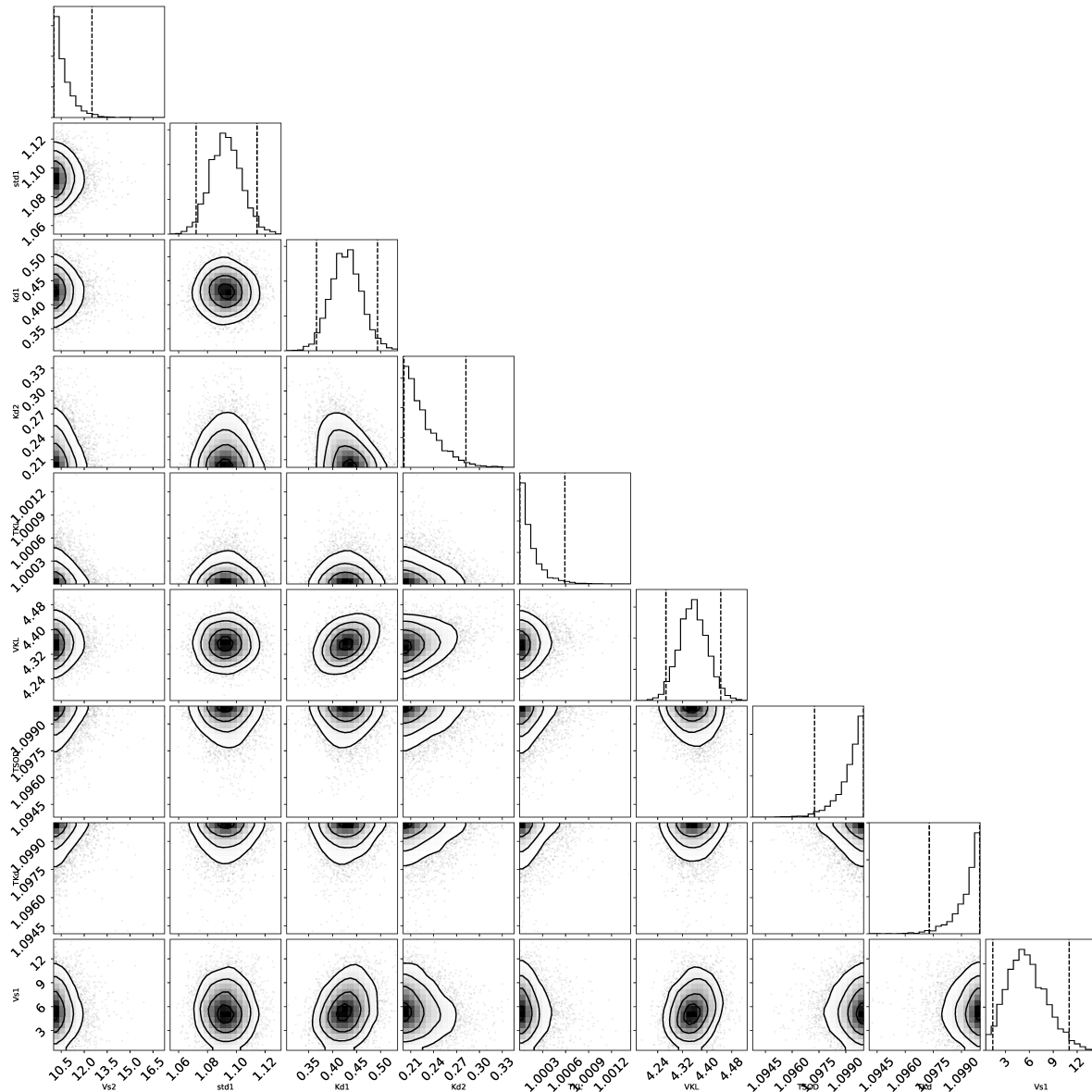


Fig 10 Posterior joint-parametric distribution for the inference of the water quality model parameters. Std1 is the hyperparameters of the selected error generation process (independent, identically distributed Gaussian).

Bayesian inference relies on the a priori definition of an error generating process (Equation 1), which constitutes the likelihood structure used to define the probability of a parameter sample being supported by the observations. The error generating process is selected based on a series of hypotheses, which can be encoded by expert guesses on the behaviour of the system. Yet those assumptions are still a subjective exercise and its validity should be checked once sampled from the posterior distribution. In this case, the initial error generation process for both flow and DO series was conceptualised as an independent, identically distributed Gaussian distribution. Once the first posterior samples for the flow inference are available it becomes apparent that the residual structure shows a mild heteroscedastic structure. This implies in this context that the residuals are positively correlated with the flow. This is well described in the hydrological literature (Sorooshian and Dracup 1980) and is corrected by using a linear dependent variance structure in the error generating process for the flow inference. Fig 11 represents three relevant characteristics of the residual structure at the posterior samples of river flow, comparing the assumed error generating process (in black) and the sampled one (in blue). Fig 11a shows the heteroscedasticity structure of the residuals, Fig 11b shows the comparison of probability density functions and Fig 11c shows the time-autocorrelation structure of the assumed and computed residuals. It is apparent that the

residual independency assumption is violated, since computed residuals present a strong time autocorrelation structure.

Fig 12 shows the comparison of the computed residuals and the assumed error generating process (independent, homoscedastic and Gaussian) for the dissolved oxygen in the river. The variance of the residuals is largely independent from the DO value. The proposed and computed probability densities have a reasonable match. Also, residuals present a clear autocorrelation structure, albeit shorter than that of the flow inference.

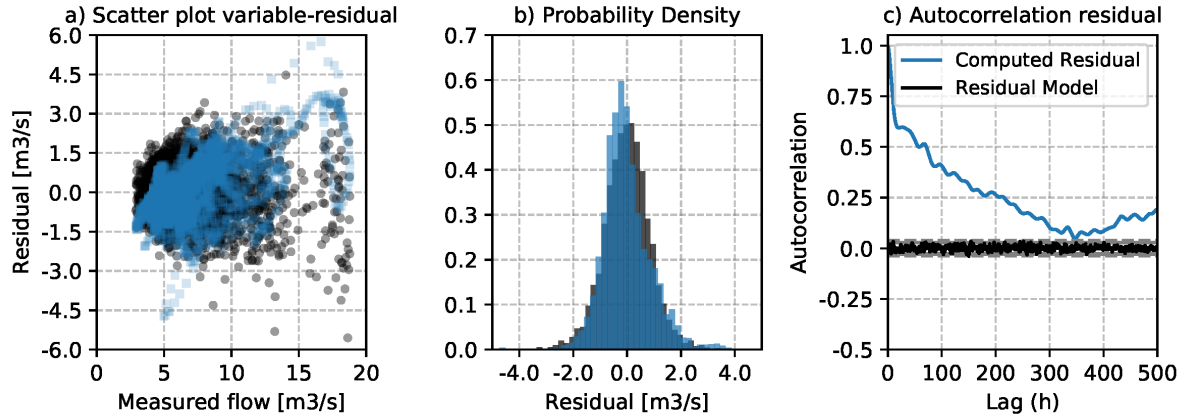


Fig 11 Residual structure at the flow posterior mean sample, a) scatter plot variable-residual showing the dependency of the variance, b) the residual probability density and c) the autocorrelation plot at different time-lags.

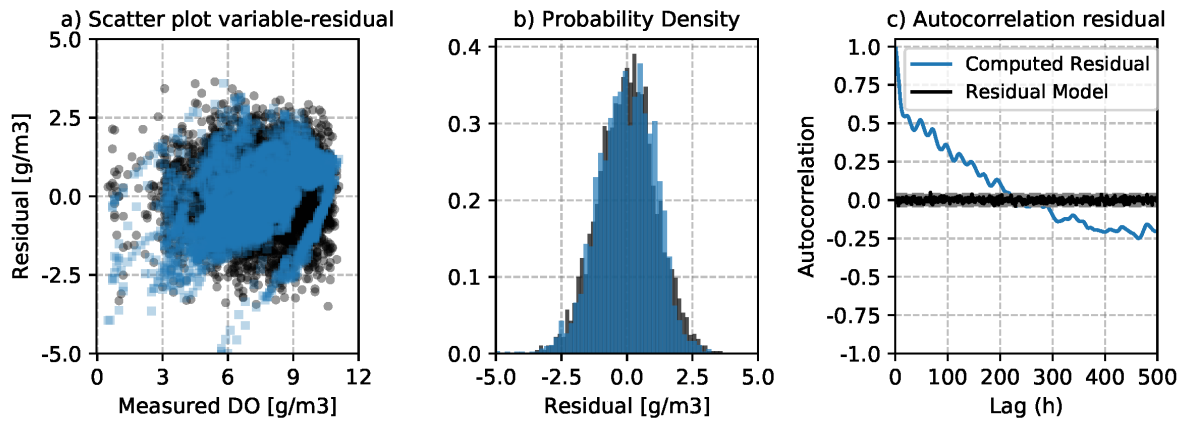


Fig 12 Residual structure at the dissolved oxygen posterior mean sample, a) scatter plot variable-residual showing the dependency of the variance, b) the residual probability density and c) the autocorrelation plot at different time-lags.

The time autocorrelation structure in hydrological inference has been discussed in several studies. For instance, Kuczera (1983) applied an ARMA (autoregressive moving-average) model to represent an autocorrelated likelihood structure in a hydrological model. Bates and Campbell (2001) argued that ARMA structures lead to local minima, and AR (autoregressive) models of order p are to be preferred. Schoups and Vrugt (2010) presented the use of an alternative likelihood structure, which addresses several common issues; like the non-normality of residuals (which was not relevant in our study), the variance non-stationarity (corrected by the use of a linear standard deviation dependency) and the temporal correlation of the residuals approached with an AR(p) model. Yet all these three studies simulated catchment hydrological flows at daily scales. In this paper, the measurement layout has an hourly time-step, since relevant processes occur at those scales. It is expected that the autocorrelation structure becomes stronger when dealing with shorter time-scales. As seen in Fig 11 and Fig 12 the correlation is still around 0.5 in the order of 50-100 hours lag time. Honti et al. (2013) and Del Giudice et

al. (2013) presented the direct encoding of a Bias description process within the error model. This was applied to urban drainage hydrodynamic simulation with time-steps of 1-2 minutes, which present strong autocorrelation structures. The Bias description can be conceptualised as function of different variables or inputs, yet in its basic form it constitutes a Gaussian multivariate distribution with an exponential covariance structure as in Equation 6.

The use of an AR(3) model in this case, rendered an almost negligible effect of the autoregressive parameters of higher order than one (<0.01), thus generating an equivalent AR(1) model. Measured series vs. inferred comparison and the residual structure can be seen at Fig 15 (Annex B). Although the autocorrelation of residuals is better represented, the fit of the mean sample did not improve, rather was degraded through accounting for the autocorrelation term. This was also discussed by (Evin et al. 2013), who showed that using AR(1) models for hydrological inference on the raw residuals can lead to strong interactions with the inferred parameters and degraded outcomes.

On the other hand, the use of a Bias description as in Del Giudice et al. (2013) becomes prohibitive for long time series. This implementation requires the inversion of a covariance matrix $\Sigma \in \mathbb{R}^{n \times n}$, being n the size of the measurement layout. In this case, hourly sampling during 15-Jan-2012 until 05-Aug-2012 has a size of 4892. Expected values of the decay parameter τ are likely to produce a highly sparse covariance matrix, thus sparse inversion optimisation could be applied (Betancourt and Alvarado 1986) yet intensive sampling for populating the posterior is still cumbersome. This large covariance matrix inversion renders the evaluation of the likelihood function computationally expensive, thus eliminating the benefits of the use of the dynamic emulator for the simulator response variables. A possible solution is to create a database of likelihood samples from the dynamic emulator which is used to build a second emulator linking the parameter space and the response of the likelihood function (Dietzel and Reichert 2014). A Gaussian process and a PCE scheme were applied to represent the likelihood surface under Bias description, however even when using 2,000 samples from the likelihood during training we had an insufficient accuracy for a reliable estimation of the likelihood surface. An illustrative example of a Bias description term is shown at Fig 16 (Annex B) in which the inference was performed in a shorter time-series (26-Jul-2012 – 14-Sep-2012). The description of the autocorrelation structure in the residuals did not allow for a better description of the process, or a better understanding of the parametric uncertainty. Ammann et al. (2018) recently studied the representation of autocorrelated likelihood structures with the conventional error models as shown here. They discussed that the use of stationary autocorrelation models deteriorates the performance of the inferred model significantly (degrading even further when increasing the measurement layout frequency). They propose that the use of non-stationary autocorrelation schemes may overcome this problem, since hydrological models are expected to lose memory under storm events (thus dry-weather and wet-weather present different residual correlation patterns). This non-stationarity could not be found in this case, being the correlation structure in dry-weather and wet-weather for short scales (0-80 hours lag) equivalent and the long structures only slightly different (Fig 13). Also, this is not expected to be applicable for DO series, where residuals are even less structured as in hydrological flow.

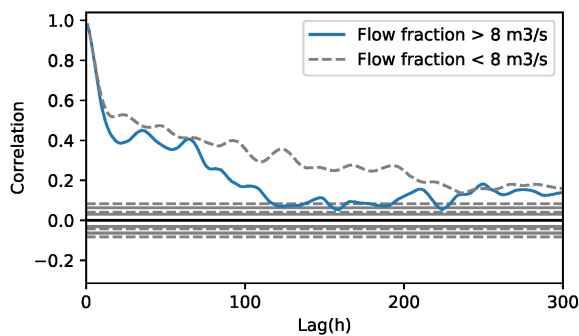


Fig 13 Autocorrelation structure for Flow residuals by magnitude

A strong autocorrelation structure is expected due to the nature of the process and the measurement layout. Both flow and dissolved oxygen concentration present several dynamic modes, induced by storm events, daily fluctuation in the WWTP effluent and variation between dry-wet periods and temperature seasons. This happens well above the measurement frequency (hourly). Small temporal shifts are expected due to model structural

misfit (e.g. incorrect CSO timing in the urban drainage scheme or misrepresentation errors in rainfall data). The temporal shift will likely render strongly correlated residuals in time. Yet these time-shifts are of little influence for the model application. The objective of the model is to represent dynamics of oxygen in a receiving water body for environmental policy assessment studies. These studies use metrics which lumps the time-dynamics, as frequency-duration-intensity tables (FWR 2012), thus the exact timing of the oxygen depletion is not highly relevant, but rather the correct representation of the magnitude and duration of the event. Therefore the use of alternative metrics (different from a formal likelihood) could further assist in the fitting of dissolved oxygen dynamics. Approaches as approximated Bayesian computation (Toni et al. 2009) should be further studied in this context.

4 CONCLUSIONS

This study presents the implementation of an integrated catchment modelling study for evaluating water quality dynamics in a large-scale urbanised system in a formal Bayesian inference scheme. The model should capture the interaction between urban drainage processes, wastewater treatment plant and receiving water dynamics. This renders complex and slow modelling platforms. Using expert-literature elicited parametric ranges for river dynamics leads to highly uncertain dissolved oxygen simulations. Thus, as a means to reduce the output variability, a local data set could be used to update prior receiving water parametric knowledge using inference schemes. This inference requires large numbers of samples. However, the computationally expensive nature of integrated catchment simulators hampers the use of sensitivity analysis or formal inference.

Two dynamic emulators (polynomial orthogonal expansions) were fitted to represent flow and dissolved oxygen depletion for a one-year long time series under four and eight global parameters respectively. Only 200 model realisations sufficed to generate an acceptable interpolation in both cases. The emulators were validated using independent data, rendering a high quality mapping between the parametric space and the dynamic response. This technique still exhibits some severe limitations, like the impossibility to include large-parametric spaces, dynamic inputs (which are not parameterised) or parametric to output non-smooth relationships (e.g. bifurcation solution points). Nevertheless, these limitations did not apply to the present study.

Both emulator schemes were used to compute the first order Sobol sensitivity indexes. Sobol indexes render a metric to assess the global sensitivity of model parameters to the output variability. This allowed increasing the understanding on the relevant processes involving flow propagation and dissolved oxygen depletion processes in the river Dommel. The sensitivity of oxygen depletion showed a non-stationary dependency across storm processes. Thus, the system does not always behave in the same manner depending on the dynamic state, the storm event and the season. In general terms, the reaeration rate showed to be the most relevant parameter under dry-weather flow. Depletion of fast-slow biodegradable matter is often the responsible for the sensitivity in the intensity of the oxygen dip (attending only to the river process parameters). Meanwhile oxygen recovery sensitivity is shared between the sediment oxygen demand and reaeration processes, with a strong curve dominated by seasonality (temperature driven).

Several hypotheses were used to construct an inference error generating process. The use of an i.i.d Normally distributed error was sufficient to represent dissolved oxygen residuals. A heteroscedastic residual structure had to be fitted to the flow propagation process. However, both inferred residual series rendered a highly temporally correlated structure, which violates the assumption of independence. The residual autocorrelation is related to the measurement layout frequency (hourly) and the nature of the simulated processes such as; a strong memory effect, model structure-induced time shifts and input estimation-measurement errors. Various formulations to deal with the residual autocorrelation or structural bias were tested. However, the inferred dynamics either deteriorated or did not improve. Detailed investigation on the effects of neglecting the correlation structure in the dissolved oxygen residual structure is still missing. The use of alternative metrics for the inference of dissolved oxygen dynamics should be further studied.

The use of a dynamic emulation scheme allows gaining insights on the underlying mechanistic relationships of the integrated urban water quality system. This can be easily extended to similar environmental modelling studies, facilitating the application of sensitivity analysis, inference or calibration under long-time series and low-dimensionality parametric spaces.

5 ACKNOWLEDGEMENTS

This research was done as part of the Marie Curie ITN—Quantifying Uncertainty in Integrated Catchment Studies (QUICS) project. This project received funding from the European Union’s Seventh Framework Programme for research, technological development and demonstration under grant agreement No.607000. The authors would like to thank MIKE DHI for a free-of-charge academic license agreement for the use of their software WEST. The authors would like to thank the financial contribution of the Dutch ‘Kennisprogramma Urban Drainage’ (Knowledge Programme Urban Drainage). The involved parties are: ARCADIS, Deltares, Evides, Gemeente Almere, Gemeente Arnhem, Gemeente Breda, Gemeente’s-Gravenhage, Gemeentewerken Rotterdam, Gemeente Utrecht, GMB Rioleringsstechniek, Grontmij, KWR Watercycle Research Institute, Royal HaskoningDHV, Stichting RIONED, STOWA, Tauw, vanderValk + deGroot, Waterboard De Dommel, Waternet and Witteveen+Bos.

6 REFERENCES

- Ammann L, Reichert P, Fenicia F (2018) A framework for likelihood functions of deterministic hydrological models *Hydrol Earth Syst Sci Discuss* 2018:1-39 doi:10.5194/hess-2018-406
- Bates BC, Campbell EP (2001) A Markov Chain Monte Carlo Scheme for parameter estimation and inference in conceptual rainfall-runoff modeling *Water Resources Research* 37:937-947 doi:doi:10.1029/2000WR900363
- Benedetti L et al. (2013) Modelling and monitoring of integrated urban wastewater systems: review on status and perspectives *Water Science and Technology* 68:1203-1215 doi:10.2166/wst.2013.397
- Betancourt R, Alvarado FL (1986) Parallel Inversion of Sparse Matrices *IEEE transactions on power systems* 1:74-81
- Carbajal JP, Leitão JP, Albert C, Rieckermann J (2017) Appraisal of data-driven and mechanistic emulators of nonlinear simulators: The case of hydrodynamic urban drainage models *Environmental Modelling & Software* 92:17-27 doi:10.1016/j.envsoft.2017.02.006
- Del Giudice D, Honti M, Scheidegger A, Albert C, Reichert P, Rieckermann J (2013) Improving uncertainty estimation in urban hydrological modeling by statistically describing bias *Hydrology and Earth System Sciences* 17:4209-4225 doi:10.5194/hess-17-4209-2013
- Deletic A et al. (2012) Assessing uncertainties in urban drainage models *Physics and Chemistry of the Earth, Parts A/B/C* 42-44:3-10 doi:10.1016/j.pce.2011.04.007
- Dietzel A, Reichert P (2014) Bayesian inference of a lake water quality model by emulating its posterior density *Water Resources Research* 50:7626-7647 doi:doi:10.1002/2012WR013086
- Dotto CB et al. (2012) Comparison of different uncertainty techniques in urban stormwater quantity and quality modelling *Water research* 46:2545-2558 doi:10.1016/j.watres.2012.02.009
- Evin G, Kavetski D, Thyer M, Kuczera G (2013) Pitfalls and improvements in the joint inference of heteroscedasticity and autocorrelation in hydrological model calibration *Water Resources Research* 49:4518-4524 doi:10.1002/wrcr.20284
- Feinberg J, Langtangen HP (2015) Chaospy: An open source tool for designing methods of uncertainty quantification *Journal of Computational Science* 11:46-57 doi:<https://doi.org/10.1016/j.jocs.2015.08.008>

-
- 467 FWR (2012) Manual, Urban Pollution Management. 3th edn.,
- 468 Gautschi W (1994) Algorithm 726: ORTHOPOL- A Package of Routines for Generating Orthogonal
469 Polynomials and Gauss-Type Quadrature Rules ACM Transactions on Mathematical
470 Software (TOMS) 20:21-62
- 471 Goodman J, Weare J (2010) Ensemble samplers with affine invariance Communications in applied mathematics
472 and computational science 5:65-80
- 473 Hadigol M, Doostan A (2018) Least squares polynomial chaos expansion: A review of sampling strategies
474 Computer Methods in Applied Mechanics and Engineering 332:382-407
475 doi:<https://doi.org/10.1016/j.cma.2017.12.019>
- 476 Hastings WK (1970) Monte Carlo sampling methods using Markov chains and their applications
- 477 Hoffman MD, Gelman A (2014) The No-U-turn sampler: adaptively setting path lengths in Hamiltonian Monte
478 Carlo Journal of Machine Learning Research 15:1593-1623
- 479 Honti M, Stamm C, Reichert P (2013) Integrated uncertainty assessment of discharge predictions with a
480 statistical error model Water Resources Research 49:4866-4884
481 doi:10.1002/wrcr.20374
- 482 Kuczera G (1983) Improved parameter inference in catchment models: 1. Evaluating parameter uncertainty
483 Water Resources Research 19:1151-1162 doi:doi:10.1029/WR019i005p01151
- 484 Laloy E, Rogiers B, Vrugt JA, Mallants D, Jacques D (2013) Efficient posterior exploration of a high-
485 dimensional groundwater model from two-stage Markov chain Monte Carlo
486 simulation and polynomial chaos expansion Water Resources Research 49:2664-2682
487 doi:10.1002/wrcr.20226
- 488 Laloy E, Vrugt JA (2012) High-dimensional posterior exploration of hydrologic models using multiple-try
489 DREAM(ZS)and high-performance computing Water Resources Research 48
490 doi:10.1029/2011wr010608
- 491 Langeveld J, Nopens I, Schilperoort R, Benedetti L, de Klein J, Amerlinck Y, Weijers S (2013a) On data
492 requirements for calibration of integrated models for urban water systems Water Sci
493 Technol 68:728-736 doi:10.2166/wst.2013.301
- 494 Langeveld J, Van Daal P, Schilperoort R, Nopens I, Flameling T, Weijers S (2017) Empirical Sewer Water
495 Quality Model for Generating Influent Data for WWTP Modelling Water 9:491
496 doi:10.3390/w9070491
- 497 Langeveld JG et al. (2013b) Impact-based integrated real-time control for improvement of the Dommel River
498 water quality Urban Water Journal 10:312-329 doi:10.1080/1573062X.2013.820332
- 499 Metropolis N, Rosenbluth AW, Rosenbluth MN, Teller AH, Teller E (1953) Equation of state calculations by
500 fast computing machines The journal of chemical physics 21:1087-1092
- 501 Moreno-Rodenas AM, Bellos V, Langeveld JG, Clemens F (2018) A dynamic emulator for physically based
502 flow simulators under varying rainfall and parametric conditions Water research
503 142:512-527 doi:10.1016/j.watres.2018.06.011

504 Moreno-Rodenas AM, Cecinati F, Langeveld J, Clemens FHLR (2017) Impact of Spatiotemporal Characteristics
505 of Rainfall Inputs on Integrated Catchment Dissolved Oxygen Simulations Water
506 9:926 doi:10.3390/w9120926

507 Muschalla D et al. (2009) The HSG procedure for modelling integrated urban wastewater systems Water Sci
508 Technol 60:2065-2075 doi:10.2166/wst.2009.576

509 Patil A, Huard D, Fonnesbeck CJ (2010) PyMC: Bayesian Stochastic Modelling in Python Journal of statistical
510 software 35:1-81

511 Rauch W, Bertrand-Krajewski J, Krebs P, Mark O, Schilling W, Schutze M, Vanrolleghem P (2002)
512 Deterministic modelling of integrated urban drainage systems Water Science &
513 Technology 45:81-94

514 Schoups G, Vrugt JA (2010) A formal likelihood function for parameter and predictive inference of hydrologic
515 models with correlated, heteroscedastic, and non-Gaussian errors Water Resources
516 Research 46:n/a-n/a doi:10.1029/2009wr008933

517 Sobol IM (1993) Sensitivity estimates for nonlinear mathematical models Mathematical modelling and
518 computational experiments 1:407-414

519 Solvi A-M (2006) Modelling the sewer-treatment-urban river system in view of the EU Water Framework
520 Directive. Doctoral Thesis, Ghent University

521 Sorooshian S, Dracup JA (1980) Stochastic parameter estimation procedures for hydrologic rainfall-runoff
522 models: Correlated and heteroscedastic error cases Water Resources Research 16:430-
523 442

524 Toni T, Welch D, Strelkowa N, Ipsen A, Stumpf MP (2009) Approximate Bayesian computation scheme for
525 parameter inference and model selection in dynamical systems Journal of the Royal
526 Society Interface 6:187-202

527 Tscheikner-Gratl F, Lepot M, Moreno-Rodenas AM, Schellart ANA (2017) QUICS - D6.7 A Framework for the
528 application of uncertainty analysis. doi:10.5281/zenodo.1240926.

529 Xiu D (2010) Numerical methods for stochastic computations: a spectral method approach. Princeton university
530 press,

531 Xiu D, Karniadakis GE (2002) The Wiener--Askey polynomial chaos for stochastic differential equations SIAM
532 journal on scientific computing 24:619-644

533 Yang J, Jakeman A, Fang G, Chen X (2018) Uncertainty analysis of a semi-distributed hydrologic model based
534 on a Gaussian Process emulator Environmental Modelling & Software 101:289-300
535 doi:<https://doi.org/10.1016/j.envsoft.2017.11.037>

536

ANNEX A: RIVER MODEL WATER QUALITY PROCESSES

Table 5 presents the process matrix for the dissolved oxygen process routine at the river section. The model accounts for the transport and conversion rates of seven concentration state variables; BOD1 (Biological oxygen demand of fast biodegradable suspended fraction), BOD2 (Biological oxygen demand of slow biodegradable suspended fraction), BOD1p (Biological oxygen demand of fast biodegradable particulate fraction), BOD2p (Biological oxygen demand of slow biodegradable particulate fraction), BODs (Biological oxygen demand at the sediment section), NH4 (Ammonium) and DO (dissolved oxygen concentration).

Table 5 Process matrix for the river water quality model structure

State Variable	BOD1	BOD1p	BOD2	BOD2p	BODs	NH4	DO	rate
	gO ₂ /m ³	gO ₂ /m ³	gO ₂ /m ³	gO ₂ /m ³	gO ₂ /m ³	gN/m ³	gO ₂ /m ³	
Process								
1a. Oxidation of fast-suspended fraction (BOD1)	-1						-1	$TKd^{T_{wat}-20} \cdot Kd1 \cdot BOD1 \cdot \frac{DO}{KO2+DO}$
1b. Oxidation of fast-particulate fraction (BOD1p)		-1					-1	$TKd^{T_{wat}-20} \cdot Kd1 \cdot BOD1p \cdot \frac{DO}{KO2+DO}$
2a. Oxidation of slow-suspended fraction (BOD2)			-1				-1	$TKd^{T_{wat}-20} \cdot Kd2 \cdot BOD2 \cdot \frac{DO}{KO2+DO}$
2b. Oxidation of slow-particulate fraction (BOD2p)				-1			-1	$TKd^{T_{wat}-20} \cdot Kd2 \cdot BOD2p \cdot \frac{DO}{KO2+DO}$
3a. Sedimentation of BOD1p		-1			+1			$Vs1 \cdot BOD1p$
3b. Sedimentation of BOD2p				-1	+1			$Vs2 \cdot BOD2p$
4. Oxidation of organic matter in the sediment					-1		-1	$TSOD^{T_{wat}-20} \cdot K_{BOD} \cdot \frac{BODs}{d} \cdot \frac{DO}{KSO+DO}$
5. Constant sediment oxygen demand							-1	$TSOD^{T_{wat}-20} \cdot \frac{SOD}{d} \cdot \frac{DO}{KSO+DO}$
6. Nitrification						-1	-4.57	$TKnit^{T_{wat}-20} \cdot Knit \cdot NH4 \cdot \frac{DO}{KN02+DO}$
7. Photosynthesis macrophyte							+1	$TKp^{T_{wat}-20} \cdot k_{prodM} \cdot Io \cdot \frac{MB}{d}$
8. Macrophyte oxygen consumption							-1	$TKp^{T_{wat}-20} \cdot k_{pcons} \cdot \frac{MB}{d}$
9. Reaeration							+1	$TKL^{T_{wat}-20} \cdot VKL \cdot (CS - DO)$

The river flow was conceptualised as a tank in series lumped scheme, in which each tank is a well-stirred reactor in which the biochemical processes take place (Fig 14). The mass balance equation is computed at each node:

$$\frac{dV_i}{dt} = Q_{in_i} - Q_{out_i}, \quad (7)$$

where V_i is the volume at tank i and Q_{in_i} the sum of inflows from CSOs, upstream sections or rural hydrology. Q_{out_i} is the outflow from the tank at each instance and is computed as a free-surface flow following the Gauckler-Manning equation:

$$Q_{out_i} = A \cdot R_H^{\frac{2}{3}} \cdot slp^{\frac{1}{2}} \cdot n^{-1}, \quad (8)$$

being A and R_H the area and hydraulic radius for a trapezoidal section with a given slope (slp) and bed Manning's roughness (n).

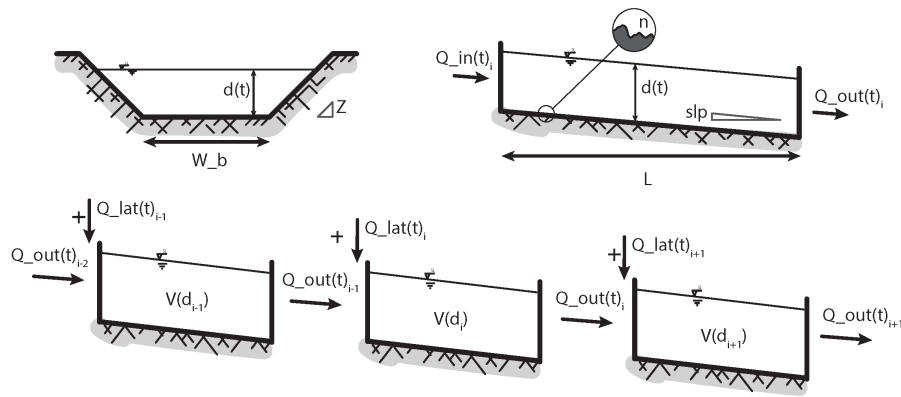


Fig 14 Tank in series river flow scheme

ANNEX B: ACCOUNTING FOR THE AUTOCORRELATION STRUCTURE

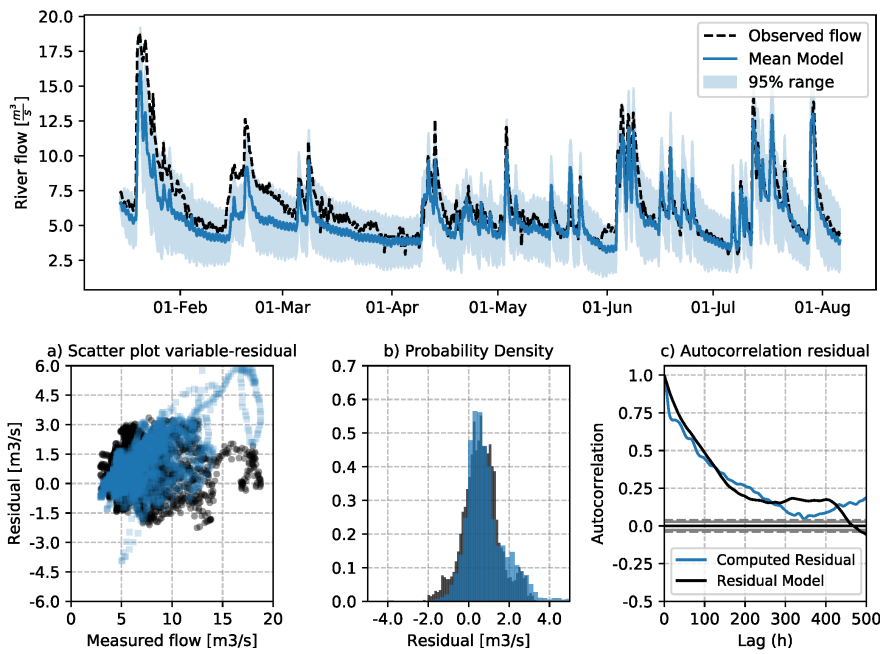


Fig 15 Autoregressive model order 3, comparison of measured and inferred dynamics and residual structure

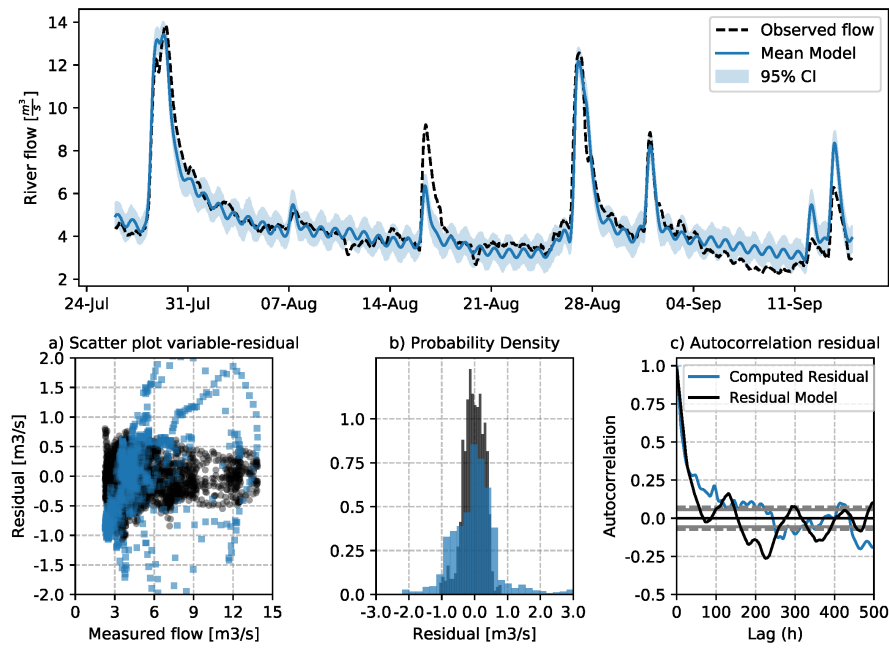


Fig 16 Bias description, comparison of measured and inferred dynamics and residual structure